

UNIT TEST REPORT

By Firmgen

Report identity

Project	vehicleparametersdisplay
Target	firmware (verified off target, no MCU hardware in the loop)
Report ID	UTR-A6BE4A98
Run ID	a6be4a98-315f-47ee-8581-50642a3b95f7
Execution date	07 November 2026, 11:25 UTC
Prepared by	Firmgen automated unit-test agent
Cases executed	145 across 40 test file(s)

1. Test Objective

This activity executed 40 test files against selected firmware units, establishing 120 passing cases out of 145 and passing evidence for 14 of 24 traced requirements. It exercised selected code associated with MPU6050 motion sensing, OLED rendering, alert configuration, application start-up, helpers and UART initialization, among other enumerated units. The results show which requirements have executed passing evidence and identify test assertions and requirements that need further work.

2. Environment and Scope

Source under test	vehicleparametersdisplay
Compile-time defines	UNIT_TEST, HOST_TEST, UNITY_INCLUDE_EXEC_TIME
Test parallelism	8 thread(s)
Test framework	Ceedling 1.1.6 (Unity + CMock)
Build system	Ceedling 1.1.6
Compiler	C:\Users\Adith\AppData\Local\craftifai\toolchains\mingw\16.2.0posix-14.0.0-ucrt-r1\bin\gcc.exe 16.2.0
Host environment	FirmGen-managed toolchain (ruby 3.3.12-1) on Windows-10
Execution mode	Off target - compiled for the build architecture, no MCU hardware in the loop; executed directly on the build host, not in a container
Test cases executed	145 across 40 test file(s)

3. Test Harness and Reproducibility

The tests were built and run with Ceedling 1.1.6, Unity and CMock, using GCC 16.2.0 with the UNIT_TEST, HOST_TEST and UNITY_INCLUDE_EXEC_TIME defines. They ran off target on the Windows-10 build host using the FirmGen-managed Ruby 3.3.12-1 toolchain, with 8-thread test parallelism; no MCU hardware was in the loop. CMock was part of the harness, and the app_start results specifically identify a missing mock return configuration for alert_indicator_update.

Reproduction step	Command	Result
Configure	`ceedling project.yml (generated per run)`	● PASS
Build and link	`ceedling test:build_only`	● PASS
Execute tests	`ceedling test:all`	● PASS

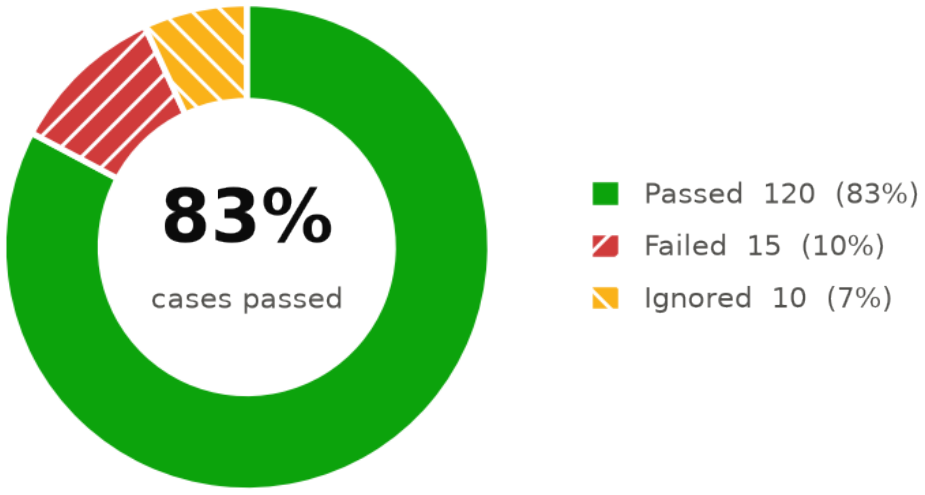
Evidence shipped with this run: executed_tests.xlsx, run.txt

4. Execution Summary

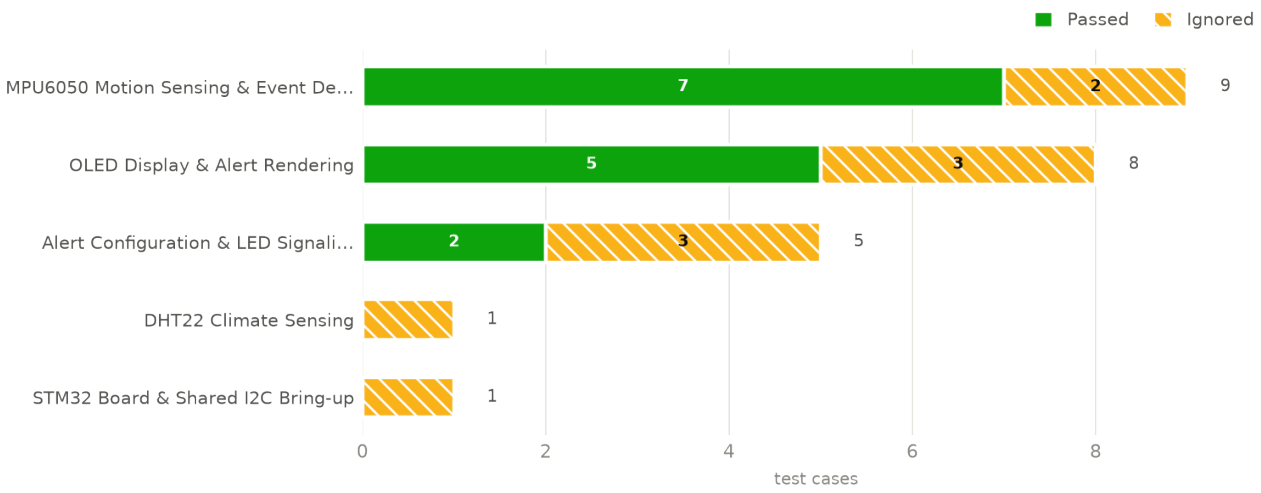
120	83%	145	15
Cases passed	Case pass rate	Cases executed	To address

Metric	Result
Cases passed	120
Case pass rate	82.8%
Requirements fully passing	14
Requirements traced to test evidence	24
Total test cases	145
Test files executed	40
Requirement-driven cases	24
Firmware-coverage cases	121
Cases awaiting a verdict	10
Cases with a failure to address	15
Requirements awaiting a verdict	10
Requirements with a failure to address	0
Total execution time	261,582 ms

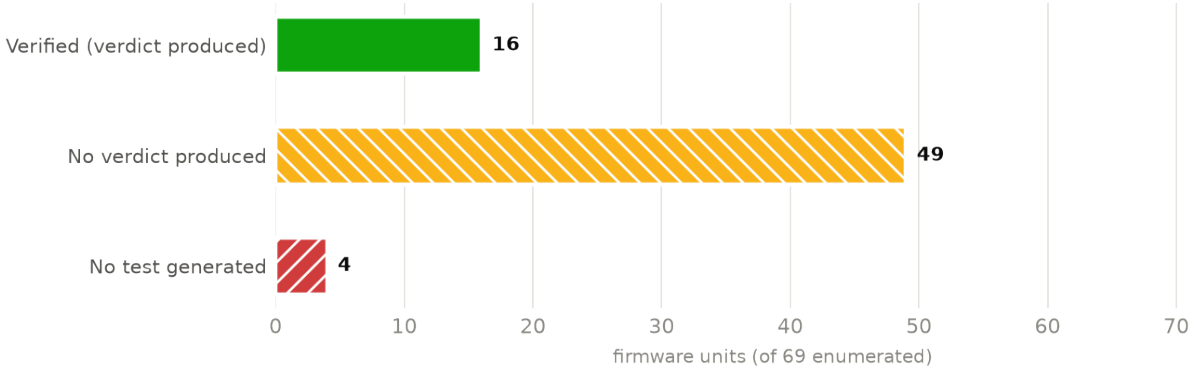
Outcome of every executed case



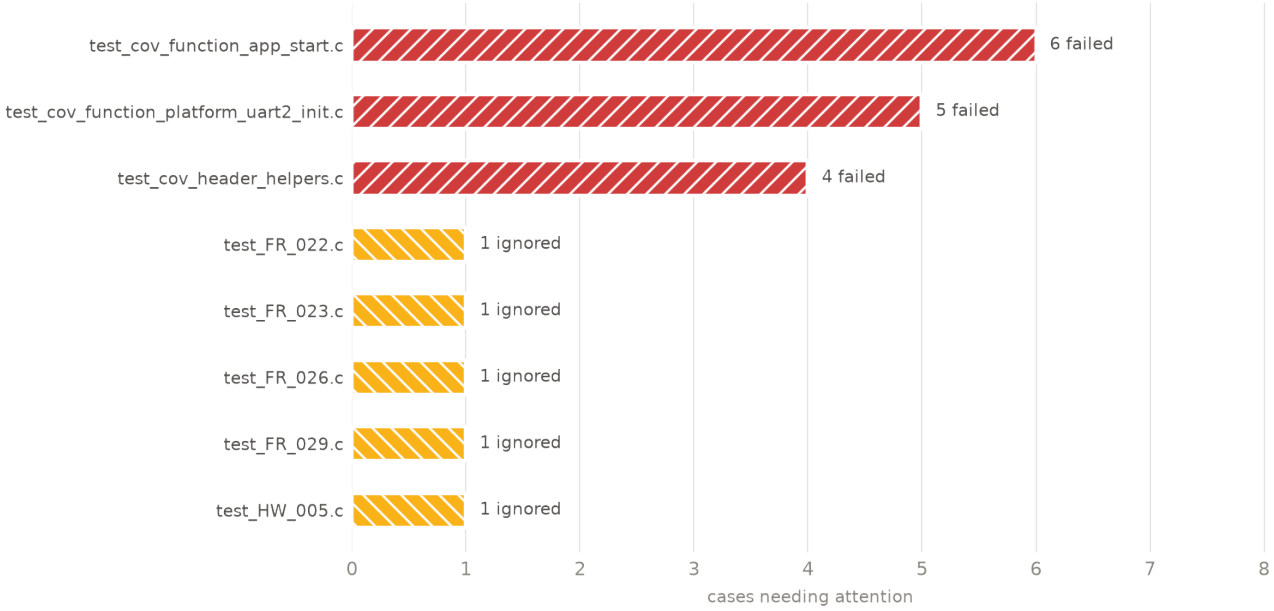
Outcomes by requirement category, strongest first



Firmware units verified by this suite



Where to focus next



5. Traceability

The executed evidence records 14 passing requirements out of 24 traced; 10 requirements are not yet verified, and none are categorized as requirements to address. The 15 cases to address concern test assertions or harness behavior, while the not-yet-verified requirements need additional observable implementation or acceptance criteria before they can be tested.

Requirements traced to executed cases

Req. ID	Category	Module / symbol	Test file	Cases	Result
FR-001	MPU6050 Motion Sensing & Event De…	app_report_motion +2 more	test_FR_001.c	1	● PASSED
FR-002	OLED Display & Alert Rendering	app_make_oled_snapshot +4 more	test_FR_002.c	1	● PASSED
FR-006	MPU6050 Motion Sensing & Event De…	motion_processor_process +1 more	test_FR_006.c	1	● PASSED
FR-008	MPU6050 Motion Sensing & Event De…	app_report_motion +2 more	test_FR_008.c	1	● PASSED
FR-010	MPU6050 Motion Sensing & Event De…	app_report_motion +2 more	test_FR_010.c	1	● PASSED
FR-011	OLED Display & Alert Rendering	app_make_oled_snapshot +4 more	test_FR_011.c	1	● PASSED
FR-012	Alert Configuration & LED Signali…	motion_processor_process	test_FR_012.c	1	● PASSED
FR-013	MPU6050 Motion Sensing & Event De…	app_report_motion +3 more	test_FR_013.c	1	● PASSED
FR-014	MPU6050 Motion Sensing & Event De…	app_report_motion +3 more	test_FR_014.c	1	● PASSED
FR-017	OLED Display & Alert Rendering	alert_indicator_update +3 more	test_FR_017.c	1	● PASSED
FR-019	OLED Display & Alert Rendering	alert_indicator_update +3 more	test_FR_019.c	1	● PASSED
FR-027	OLED Display & Alert Rendering	app_make_oled_snapshot +4 more	test_FR_027.c	1	● PASSED
FR-036	MPU6050 Motion Sensing & Event De…	app_report_motion +3 more	test_FR_036.c	1	● PASSED
FR-037	Alert Configuration & LED Signali…	alert_indicator_init +4 more	test_FR_037.c	1	● PASSED
FR-022	Alert Configuration & LED Signali…	motion_processor_process	test_FR_022.c	1	● IGNORED
FR-023	Alert Configuration & LED Signali…	motion_processor_process	test_FR_023.c	1	● IGNORED
FR-026	Alert Configuration & LED Signali…	motion_processor_process	test_FR_026.c	1	● IGNORED
FR-029	OLED Display & Alert Rendering	app_make_oled_snapshot +4 more	test_FR_029.c	1	● IGNORED

Req. ID	Category	Module / symbol	Test file	Cases	Result
HW-005	STM32 Board & Shared I2C Bring-up	PRD.md	test_HW_005.c	1	IGNORED
IF-001	DHT22 Climate Sensing	HAL_TIM_Base_MspInit +1 more	test_IF_001.c	1	IGNORED
IF-004	MPU6050 Motion Sensing & Event De...	PRD.md	test_IF_004.c	1	IGNORED
IF-005	MPU6050 Motion Sensing & Event De...	PRD.md	test_IF_005.c	1	IGNORED
IF-006	OLED Display & Alert Rendering	PRD.md	test_IF_006.c	1	IGNORED
IF-007	OLED Display & Alert Rendering	PRD.md	test_IF_007.c	1	IGNORED

Firmware units traced to executed cases

Firmware unit	Kind	Test file	Cases	Result
alert_indicator	module	test_cov_module_alert_indicator.c	6	PASSED
board	header	test_cov_header_board.c	2	PASSED
dht22_decoder	module	test_cov_module_dht22_decoder.c	5	PASSED
firmgen_telemetry_start	function	test_cov_function_firmgen_telemetry_start.c	1	PASSED
logger_platform_write	function	test_cov_function_logger_platform_write.c	4	PASSED
motion_processor	module	test_cov_module_motion_processor.c	24	PASSED
motion_processor_event_name	function	test_cov_function_motion_processor_event_name.c	6	PASSED
motion_service_init	function	test_cov_function_motion_service_init.c	4	PASSED
motion_service_poll	function	test_cov_function_motion_service_poll.c	12	PASSED
mpu6050	module	test_cov_module_mpu6050.c	10	PASSED
oled_port_write	function	test_cov_function_oled_port_write.c	13	PASSED
platform_uart2_write	function	test_cov_function_platform_uart2_write.c	4	PASSED
sh1106_clear	function	test_cov_function_sh1106_clear.c	3	PASSED
app_start	function	test_cov_function_app_start.c	6	FAILED
helpers	header	test_cov_header_helpers.c	15	FAILED
platform_uart2_init	function	test_cov_function_platform_uart2_init.c	6	FAILED

6. Failure, Ignored Case, and Root-Cause Record

Scope / cases	Observed result	Suspected cause	Severity	Disposition
configuration requirements without observable configuration… FR-022, FR-023, FR-026	These cases were ignored because the supplied scope does not expose the temperature or humidity limits needed for testing, and the sheet records no acceptance criterion for these requirements.	The relevant configuration behavior is not observable in the supplied code and the requirements lack recorded acceptance criteria.	● Low	Deferred; define acceptance criteria and provide an observable implementation or interface for the limits before adding executable tests.
climate rendering implementation not supplied FR-029	The case was ignored because the implementation of oled_service_draw_climate, which determines humidity text and formatting, is not present in the supplied source; the sheet records no acceptance criterion.	The implementation needed to inspect the behavior is outside the supplied source scope, and the requirement has no recorded acceptance criterion.	● Low	Deferred; provide the implementation and define the acceptance criterion, then test the rendered humidity text and formatting.
physical wiring and power requirements not observable in su… HW-005, IF-001, IF-004, IF-005, IF-006, IF-007	These cases were ignored because the supplied material does not represent the specified board power path or sensor and display supply and ground connections; the sheet records no acceptance criterion for these requirements.	The requirements concern physical connections that cannot be determined from the supplied firmware scope.	● Low	Deferred; define acceptance criteria and verify these connections using suitable board documentation or hardware-level evidence.
app_start mock return configuration no ids cited	Each case reports that alert_indicator_update requires _IgnoreAndReturn.	The CMock expectation for alert_indicator_update does not configure a return value for these cases.	● Medium	Open; configure an explicit mock return or an appropriate ignore-and-return behavior in each case, then rerun the tests.
helper assertion mismatches no ids cited	The four assertions report expected and observed values of 4 and 2, 2 and 3, 6 and 7, and 1 and 4294967295, respectively.	The asserted expectations differ from the helper results; the intended behavior, including the unsigned boundary case, needs to be checked against the implementation and test inputs.	● Medium	Open; review the helper behavior and boundary expectations, update the implementation or assertions as appropriate, and rerun these cases.

Scope / cases	Observed result	Suspected cause	Severity	Disposition
platform_uart2_init assertion mismatches no ids cited	All five cases expected 1 and observed 0.	The expected initialization result differs from the observed result; the initialization path and any mocked peripheral setup need review against the intended behavior.	● Medium	Open; check the UART initialization behavior and test setup, then align the assertion or implementation with the intended result and rerun.

7. Limitations and Exclusions

This report states what the executed cases established. The boundaries below mark where off-target unit testing ends and other verification activities begin - they are the scope of this activity, not defects in it.

- These off-target host tests do not establish electrical behavior or physical wiring; board-level inspection and hardware testing cover those properties.
- This activity does not establish interrupt latency; timing-focused verification on the target is needed for that evidence.
- Host unit tests do not establish RTOS scheduling behavior; system-level testing on the target is needed for that evidence.
- The activity verified 16 of 69 enumerated firmware units, so it does not establish behavior for the remaining units.
- The results apply to the supplied source and test scope and do not establish behavior of implementations that were not present.

8. Verification Status and Conclusion

Decision field	Conclusion
Verified scope	120 of 145 executed case(s) passed, covering 14 requirement(s) in full across 40 test file(s).
Outstanding items	Next, configure the app_start mock returns, investigate and resolve the helper and platform_uart2_init assertion mismatches, and rerun the affected cases. For the 10 requirements not yet verified, provide the missing implementation or observable evidence and acceptance criteria where needed, then a…
Conclusion	The executed suite provides passing evidence for 120 cases, 14 traced requirements and 16 enumerated firmware units, while also identifying 15 cases with assertions or mock setup to address. The remaining unverified requirements are tied to unavailable implementation, physical connection evidence or missing acceptance criteria, rather than being recorded as requirements to address. Completing the test and evidence work described above will extend traceability without drawing conclusions beyond this host-test activity.

Generated automatically from the executed test results of run a6be4a98-315f-47ee-8581-50642a3b95f7. The figures in this document are computed from executed_tests.xlsx; no count in it was written by a language model.

9. Appendix: Complete Case Record

Every case this run executed appears below, one row each - 145 in total, R1-R24 from the requirement suite and C1-C121 from the firmware-coverage suite. These are the rows every figure in Section 4 was counted from.

Case	Identifier	Test function	Result
R1	FR-001	test_FR_001_01	PASSED
R2	FR-002	test_FR_002_01	PASSED
R3	FR-006	test_FR_006_01	PASSED
R4	FR-008	test_FR_008_01	PASSED
R5	FR-010	test_FR_010_01	PASSED
R6	FR-011	test_FR_011_01	PASSED
R7	FR-012	test_FR_012_01	PASSED
R8	FR-013	test_FR_013_01	PASSED
R9	FR-014	test_FR_014_01	PASSED
R10	FR-017	test_FR_017_01	PASSED
R11	FR-019	test_FR_019_01	PASSED
R12	FR-022	test_FR_022_01	IGNORED
R13	FR-023	test_FR_023_01	IGNORED
R14	FR-026	test_FR_026_01	IGNORED
R15	FR-027	test_FR_027_01	PASSED
R16	FR-029	test_FR_029_01	IGNORED
R17	FR-036	test_FR_036_01	PASSED
R18	FR-037	test_FR_037_01	PASSED
R19	HW-005	test_HW_005_01	IGNORED
R20	IF-001	test_IF_001_01	IGNORED
R21	IF-004	test_IF_004_01	IGNORED
R22	IF-005	test_IF_005_01	IGNORED
R23	IF-006	test_IF_006_01	IGNORED
R24	IF-007	test_IF_007_01	IGNORED
C1	alert_indicator	test_alert_indicator_01	PASSED
C2	alert_indicator	test_alert_indicator_02	PASSED
C3	alert_indicator	test_alert_indicator_03	PASSED
C4	alert_indicator	test_alert_indicator_04	PASSED
C5	alert_indicator	test_alert_indicator_05	PASSED
C6	alert_indicator	test_alert_indicator_06	PASSED
C7	app_start	test_app_start_01	FAILED
C8	app_start	test_app_start_02	FAILED
C9	app_start	test_app_start_03	FAILED
C10	app_start	test_app_start_04	FAILED
C11	app_start	test_app_start_05	FAILED

Case	Identifier	Test function	Result
C12	app_start	test_app_start_06	FAILED
C13	board	test_board_01	PASSED
C14	board	test_board_02	PASSED
C15	dht22_decoder	test_dht22_decoder_01	PASSED
C16	dht22_decoder	test_dht22_decoder_02	PASSED
C17	dht22_decoder	test_dht22_decoder_03	PASSED
C18	dht22_decoder	test_dht22_decoder_04	PASSED
C19	dht22_decoder	test_dht22_decoder_05	PASSED
C20	firmgen_telemetry_start	test_firmgen_telemetry_start_01	PASSED
C21	helpers	test_helpers_01	PASSED
C22	helpers	test_helpers_02	PASSED
C23	helpers	test_helpers_03	PASSED
C24	helpers	test_helpers_04	PASSED
C25	helpers	test_helpers_05	FAILED
C26	helpers	test_helpers_06	PASSED
C27	helpers	test_helpers_07	FAILED
C28	helpers	test_helpers_08	PASSED
C29	helpers	test_helpers_09	PASSED
C30	helpers	test_helpers_10	PASSED
C31	helpers	test_helpers_11	PASSED
C32	helpers	test_helpers_12	PASSED
C33	helpers	test_helpers_13	PASSED
C34	helpers	test_helpers_14	FAILED
C35	helpers	test_helpers_15	FAILED
C36	logger_platform_write	test_logger_platform_write_01	PASSED
C37	logger_platform_write	test_logger_platform_write_02	PASSED
C38	logger_platform_write	test_logger_platform_write_03	PASSED
C39	logger_platform_write	test_logger_platform_write_04	PASSED
C40	motion_processor	test_motion_processor_01	PASSED
C41	motion_processor	test_motion_processor_02	PASSED
C42	motion_processor	test_motion_processor_03	PASSED
C43	motion_processor	test_motion_processor_04	PASSED
C44	motion_processor	test_motion_processor_05	PASSED
C45	motion_processor	test_motion_processor_06	PASSED
C46	motion_processor	test_motion_processor_07	PASSED
C47	motion_processor	test_motion_processor_08	PASSED
C48	motion_processor	test_motion_processor_09	PASSED
C49	motion_processor	test_motion_processor_10	PASSED
C50	motion_processor	test_motion_processor_11	PASSED

Case	Identifier	Test function	Result
C51	motion_processor	test_motion_processor_12	● PASSED
C52	motion_processor	test_motion_processor_13	● PASSED
C53	motion_processor	test_motion_processor_14	● PASSED
C54	motion_processor	test_motion_processor_15	● PASSED
C55	motion_processor	test_motion_processor_16	● PASSED
C56	motion_processor	test_motion_processor_17	● PASSED
C57	motion_processor	test_motion_processor_18	● PASSED
C58	motion_processor	test_motion_processor_19	● PASSED
C59	motion_processor	test_motion_processor_20	● PASSED
C60	motion_processor	test_motion_processor_21	● PASSED
C61	motion_processor	test_motion_processor_22	● PASSED
C62	motion_processor	test_motion_processor_23	● PASSED
C63	motion_processor	test_motion_processor_24	● PASSED
C64	motion_processor_event_name	test_motion_processor_event_name_01	● PASSED
C65	motion_processor_event_name	test_motion_processor_event_name_02	● PASSED
C66	motion_processor_event_name	test_motion_processor_event_name_03	● PASSED
C67	motion_processor_event_name	test_motion_processor_event_name_04	● PASSED
C68	motion_processor_event_name	test_motion_processor_event_name_05	● PASSED
C69	motion_processor_event_name	test_motion_processor_event_name_06	● PASSED
C70	motion_service_init	test_motion_service_init_01	● PASSED
C71	motion_service_init	test_motion_service_init_02	● PASSED
C72	motion_service_init	test_motion_service_init_03	● PASSED
C73	motion_service_init	test_motion_service_init_04	● PASSED
C74	motion_service_poll	test_motion_service_poll_01	● PASSED
C75	motion_service_poll	test_motion_service_poll_02	● PASSED
C76	motion_service_poll	test_motion_service_poll_03	● PASSED
C77	motion_service_poll	test_motion_service_poll_04	● PASSED
C78	motion_service_poll	test_motion_service_poll_05	● PASSED
C79	motion_service_poll	test_motion_service_poll_06	● PASSED
C80	motion_service_poll	test_motion_service_poll_07	● PASSED
C81	motion_service_poll	test_motion_service_poll_08	● PASSED
C82	motion_service_poll	test_motion_service_poll_09	● PASSED
C83	motion_service_poll	test_motion_service_poll_10	● PASSED
C84	motion_service_poll	test_motion_service_poll_11	● PASSED
C85	motion_service_poll	test_motion_service_poll_12	● PASSED
C86	mpu6050	test_mpu6050_01	● PASSED
C87	mpu6050	test_mpu6050_02	● PASSED
C88	mpu6050	test_mpu6050_03	● PASSED
C89	mpu6050	test_mpu6050_04	● PASSED

Case	Identifier	Test function	Result
C90	mpu6050	test_mpu6050_05	PASSED
C91	mpu6050	test_mpu6050_06	PASSED
C92	mpu6050	test_mpu6050_07	PASSED
C93	mpu6050	test_mpu6050_08	PASSED
C94	mpu6050	test_mpu6050_09	PASSED
C95	mpu6050	test_mpu6050_10	PASSED
C96	oled_port_write	test_oled_port_write_01	PASSED
C97	oled_port_write	test_oled_port_write_02	PASSED
C98	oled_port_write	test_oled_port_write_03	PASSED
C99	oled_port_write	test_oled_port_write_04	PASSED
C100	oled_port_write	test_oled_port_write_05	PASSED
C101	oled_port_write	test_oled_port_write_06	PASSED
C102	oled_port_write	test_oled_port_write_07	PASSED
C103	oled_port_write	test_oled_port_write_08	PASSED
C104	oled_port_write	test_oled_port_write_09	PASSED
C105	oled_port_write	test_oled_port_write_10	PASSED
C106	oled_port_write	test_oled_port_write_11	PASSED
C107	oled_port_write	test_oled_port_write_12	PASSED
C108	oled_port_write	test_oled_port_write_13	PASSED
C109	platform_uart2_init	test_platform_uart2_init_01	FAILED
C110	platform_uart2_init	test_platform_uart2_init_02	FAILED
C111	platform_uart2_init	test_platform_uart2_init_03	FAILED
C112	platform_uart2_init	test_platform_uart2_init_04	FAILED
C113	platform_uart2_init	test_platform_uart2_init_05	FAILED
C114	platform_uart2_init	test_platform_uart2_init_06	PASSED
C115	platform_uart2_write	test_platform_uart2_write_01	PASSED
C116	platform_uart2_write	test_platform_uart2_write_02	PASSED
C117	platform_uart2_write	test_platform_uart2_write_03	PASSED
C118	platform_uart2_write	test_platform_uart2_write_04	PASSED
C119	sh1106_clear	test_sh1106_clear_01	PASSED
C120	sh1106_clear	test_sh1106_clear_02	PASSED
C121	sh1106_clear	test_sh1106_clear_03	PASSED